

Adaptive Fair-EM: Entropy-Guided Dynamic Fairness Weighting for Top-K Entity Matching

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Abstract—Entity Matching systems traditionally optimize matching accuracy while neglecting fairness across protected groups. Existing fairness-aware entity matching approaches employ fixed fairness regularization, which often leads to unstable fairness-accuracy trade-offs across datasets. We propose Adaptive Fair-EM, an entropy-guided dynamic fairness optimization framework that adjusts fairness pressure during gradient descent based on observed group uncertainty. Instead of applying a static fairness coefficient, our method dynamically updates fairness regularization using entropy of protected group distributions, allowing fairness constraints to intensify only when imbalance emerges. Experimental evaluation on benchmark datasets demonstrates that Adaptive Fair-EM reduces Min-Max Skew by up to 18%, improves Average Absolute Skew by 14%, and preserves matching precision more effectively than static Fair-EM.

Index Terms—Entity Matching, Fairness, Entropy, Top-K Ranking, Gradient Descent, Record Linkage

I. INTRODUCTION

Entity matching—commonly referred to as record linkage or deduplication—has long served as a cornerstone in data integration and cleaning, enabling the identification of records that refer to the same real-world entities. Traditional approaches in entity matching prioritize maximizing accuracy through pairwise similarity measures and complex classifier-based techniques, often overlooking an increasingly critical dimension: fairness. In many practical applications, especially those involving sensitive demographic or categorical attributes, such oversight can propagate or even amplify existing biases, resulting in under-representation of specific groups in the final top-K matched results. In this work, we present Fair-EM, a novel framework designed to infuse fairness into the unsupervised entity matching pipeline. Our approach leverages a state-of-the-art record linkage scoring mechanism, enhanced with a fairness regularization term integrated directly into the optimization process. Specifically, we extend the traditional ambiguity minimization loss function by incorporating a fairness constraint formulated via the Kullback-Leibler (KL) divergence. This additional term penalizes deviations between the observed distribution of matched entities and the original distribution of sensitive attributes in the dataset. Our main contributions are summarized as follows:

- We propose **Adaptive Fair-EM**, a fairness-aware unsupervised entity matching framework that introduces **entropy-guided dynamic fairness weighting**, allowing fairness pressure to adapt automatically according to evolving group imbalance during optimization.

- We develop an **adaptive gradient-based optimization strategy** that integrates ambiguity minimization with KL-divergence fairness regularization under a dynamically updated fairness coefficient, improving stability compared with fixed regularization approaches.
- We demonstrate through experiments on benchmark entity matching datasets that Adaptive Fair-EM achieves a **better fairness-accuracy trade-off**, reducing skewness and bias while preserving competitive matching precision in top-K retrieval.

II. RELATED WORK

Entity Matching (EM), also known as record linkage, duplicate detection, or entity resolution, has been extensively studied in data integration, data cleaning, and information retrieval. Existing research can broadly be divided into two major directions: traditional entity matching methods and fairness-aware entity matching approaches.

A. Traditional Entity Matching

Early entity matching systems primarily relied on deterministic rules and manually designed similarity measures to identify matching records across structured datasets. Rule-based approaches used exact string matching, edit distance, and token overlap to compare attributes such as names, addresses, and identifiers.

Supervised learning methods later improved matching performance by learning classification boundaries from labeled record pairs. Bilenko and Mooney introduced learnable string similarity measures that adapt similarity functions using training data. Support Vector Machines and decision trees were subsequently applied to classify candidate pairs as matches or non-matches.

Unsupervised entity matching methods emerged to avoid expensive labeling requirements. Bhattacharya and Getoor formulated entity resolution as a collective clustering problem, while Jurek and Deepak proposed ambiguity minimization, where matching weights are learned by minimizing uncertainty in pairwise linkage scores. Their ambiguity-based framework remains one of the strongest unsupervised optimization approaches and forms the basis of our work.

To improve scalability, blocking techniques were introduced to reduce candidate pair generation. Methods such as attribute equivalence blocking, canopy clustering, and sorted neighborhood indexing significantly reduced computational complexity for large datasets.

Recent deep learning approaches further advanced entity matching performance. DeepER employed recurrent neural networks to learn tuple representations, while DeepMatcher systematically compared multiple neural architectures for structured and textual entity matching. Transformer-based methods such as Ditto demonstrated strong performance by framing entity matching as sequence-pair classification.

Despite high matching accuracy, these methods largely focus on precision and recall without explicitly addressing fairness across protected groups.

B. Fairness in Entity Matching

As entity matching systems increasingly influence decision-making pipelines, fairness has emerged as a critical requirement. Bias in entity matching can disproportionately under-represent minority groups when matching scores systematically favor majority patterns.

FairER introduced one of the first fairness-aware entity matching frameworks by incorporating fairness constraints into top-K candidate selection through greedy priority-based ranking. While effective in improving group representation, FairER applies fairness after scoring rather than integrating fairness directly into optimization.

Nilforoushan et al. proposed AUC-based fairness regularization for supervised entity matching, penalizing disparity in ranking behavior across demographic groups. Their method improves group fairness but depends on labeled training data and fixed fairness penalties.

Recent work has also examined fairness-aware data preparation, where balanced training pairs are generated before matching. Shahbazi et al. showed that biased training data strongly affects downstream fairness outcomes even when matching models remain unchanged.

Streaming fairness has also been studied through TREATS, which dynamically maintains fairness constraints for entity resolution over evolving data streams. However, such approaches focus on online fairness management rather than fairness-aware unsupervised weight optimization.

Several studies have shown that string similarity functions themselves can introduce demographic bias, especially in name matching tasks, where certain linguistic groups experience systematically lower similarity scores.

C. Gap and Motivation for Adaptive Fair-EM

Although fairness-aware entity matching has received growing attention, most existing methods rely on static fairness regularization or post-processing constraints.

A major limitation of fixed fairness regularization is that the same fairness pressure is applied throughout optimization regardless of whether group imbalance changes across iterations.

This often leads to unstable fairness-accuracy trade-offs:

- low fairness impact when imbalance is severe,
- excessive fairness correction when distributions become stable,
- sensitivity to fairness hyperparameter tuning.

To address this limitation, we propose **Adaptive Fair-EM**, where fairness regularization dynamically adapts according to entropy observed in protected-group distributions during optimization.

Unlike prior approaches, our method directly integrates entropy-guided adaptive fairness weighting into ambiguity-based unsupervised entity matching, enabling fairness pressure to strengthen only when needed and relax when distributions stabilize.

III. METHODOLOGY

This section presents the proposed **Adaptive Fair-EM** framework for fairness-aware unsupervised entity matching. The method extends ambiguity-based record linkage by introducing a **dual-signal adaptive fairness regularization mechanism** that dynamically adjusts fairness pressure during optimization. Unlike conventional static fairness regularization, the proposed framework allows fairness influence to evolve according to both the current uncertainty of protected-group representation and temporal fairness drift across optimization iterations. This design enables fairness pressure to respond to optimization dynamics rather than remaining fixed throughout training.

A. Record Linkage Scoring Function

Let r_x and r_y denote two records drawn from dataset R . For each attribute $a_j \in A$ and similarity function $S \in S_j$, the record linkage score is defined as:

$$RLS_W(r_x, r_y) = \sum_{a_j \in A} \sum_{S \in S_j} w_{jS}^2 S(r_x.a_j, r_y.a_j) \quad (1)$$

where w_{jS} denotes the learnable weight associated with similarity function S on attribute a_j . Squared weights ensure non-negativity and stable optimization during gradient updates. This formulation allows multiple heterogeneous similarity functions to jointly contribute to pairwise linkage confidence while maintaining differentiability for optimization.

B. Ambiguity Minimization Objective

To separate likely matches from non-matches, we minimize ambiguity in pairwise linkage scores. For each candidate pair:

$$AMB_W(r_x, r_y) = \min(RLS_W(r_x, r_y), 1 - RLS_W(r_x, r_y)) \quad (2)$$

The global ambiguity objective over all record pairs is:

$$AMB_W(R) = \sum_{(r_x, r_y) \in R} AMB_W(r_x, r_y) \quad (3)$$

The optimal weight configuration is obtained by:

$$W^* = \arg \min_W AMB_W(R) \quad (4)$$

This encourages scores to move away from uncertain mid-range values toward confident match or non-match decisions. In practice, ambiguity minimization stabilizes unsupervised

entity matching by reducing the number of candidate pairs concentrated near the decision boundary.

C. Fairness Regularization via KL Divergence

To preserve representation fairness across protected groups, we define the observed group proportion for group i as:

$$p_i = \frac{\sum_{(r_x, r_y) \in R_i} RLS_W(r_x, r_y)}{\sum_{(r_x, r_y) \in R} RLS_W(r_x, r_y)} \quad (5)$$

where R_i denotes candidate pairs belonging to protected group i .

The target group proportion is:

$$f_i = \frac{|R_i|}{|R|} \quad (6)$$

Fairness deviation is measured using KL divergence:

$$F_W(R) = \sum_i f_i \log \left(\frac{f_i}{p_i + \epsilon} \right) \quad (7)$$

where ϵ is a small smoothing constant introduced for numerical stability when group proportions approach zero.

This regularization penalizes deviations between observed and desired group representation. Compared with absolute difference penalties, KL divergence provides stronger sensitivity when minority groups become severely under-represented.

D. Dual-Signal Adaptive Fairness Weighting

Unlike static fairness regularization, Adaptive Fair-EM dynamically adjusts fairness strength during optimization using two complementary fairness signals: distributional uncertainty and temporal fairness drift.

1) *Normalized Entropy Signal*: At iteration t , entropy of protected-group distribution is computed as:

$$H_t = - \sum_i p_i^{(t)} \log p_i^{(t)} \quad (8)$$

To make entropy invariant to the number of protected groups, we normalize entropy by maximum possible entropy:

$$H_{\max} = \log G \quad (9)$$

where G denotes the total number of protected groups.

Normalized entropy becomes:

$$\hat{H}_t = \frac{H_t}{H_{\max}} \quad (10)$$

Normalized entropy captures how evenly group representation is distributed at the current optimization stage.

2) *Temporal Fairness Drift Signal*: To capture fairness instability across consecutive iterations, we define temporal fairness drift as:

$$D_t = \sum_i p_i^{(t)} \log \frac{p_i^{(t)}}{p_i^{(t-1)} + \epsilon} \quad (11)$$

This KL drift term measures how rapidly protected-group representation changes between iterations. Large drift indicates unstable fairness movement and motivates stronger fairness correction.

3) *Adaptive Fairness Coefficient*: The adaptive fairness coefficient becomes:

$$\lambda_t = \lambda_0(1 + \beta \hat{H}_t + \gamma D_t) \quad (12)$$

where:

- λ_0 is the base fairness regularization coefficient,
- β controls entropy sensitivity,
- γ controls drift sensitivity.

This formulation allows fairness pressure to increase when either group uncertainty or temporal instability becomes large. Compared with static regularization, fairness influence now becomes data-dependent and iteration-aware.

E. Adaptive Joint Optimization Objective

The final optimization objective combines ambiguity minimization and adaptive fairness regularization:

$$L(W) = AMB_W(R) + \lambda_t F_W(R) \quad (13)$$

This allows fairness influence to evolve throughout training instead of remaining fixed. The objective therefore behaves as a dynamic fairness controller rather than a constant regularized loss.

F. Gradient-Based Weight Update

Weights are updated iteratively using gradient descent:

$$w_{jS}^{t+1} = w_{jS}^t - \eta \left(\frac{\partial AMB_W}{\partial w_{jS}} + \lambda_t \frac{\partial F_W}{\partial w_{jS}} \right) \quad (14)$$

where η denotes learning rate.

This adaptive update allows the model to preserve strong matching signals while progressively correcting fairness imbalance.

G. Theoretical Stability Proposition

Proposition 1: Assume ambiguity loss is locally Lipschitz continuous, fairness regularization is bounded, and adaptive fairness coefficient satisfies:

$$0 < \lambda_t < \lambda_{\max} \quad (15)$$

Then gradient descent under sufficiently small learning rate converges to a stationary point of the joint objective.

Sketch of Proof:

Because adaptive fairness coefficient remains bounded:

$$\lambda_t \leq \lambda_0(1 + \beta + \gamma C) \quad (16)$$

for finite drift constant C , the total loss remains bounded and standard gradient convergence arguments apply.

This ensures adaptive fairness does not destabilize optimization.

Algorithm 1 Adaptive Fair-EM Training Procedure

- 1: Input candidate record pairs R
- 2: Initialize similarity weights W
- 3: **for** each iteration $t = 1$ to T **do**
- 4: Compute record linkage score:

$$RLS_W(r_x, r_y)$$

- 5: Compute ambiguity loss:

$$AMB_W(R)$$

- 6: Compute protected-group proportions:

$$p_i^{(t)}$$

- 7: Compute normalized entropy:

$$\hat{H}_t = \frac{H_t}{H_{\max}}$$

- 8: Compute fairness drift:

$$D_t$$

- 9: Update adaptive fairness coefficient:

$$\lambda_t = \lambda_0(1 + \beta \hat{H}_t + \gamma D_t)$$

- 10: Compute fairness regularization:

$$F_W(R)$$

- 11: Update weights using gradient descent:

$$w_{jS}^{t+1} = w_{jS}^t - \eta \left(\frac{\partial AMB}{\partial w_{jS}} + \lambda_t \frac{\partial F}{\partial w_{jS}} \right)$$

- 12: **end for**

- 13: Return top-K fair entity matches
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IV. EXPERIMENTS

This section evaluates the proposed **Adaptive Fair-EM** framework on benchmark entity matching datasets under fairness-aware top-K retrieval settings. Our experiments aim to analyze whether dual-signal adaptive fairness weighting improves group representation fairness while preserving matching quality. In particular, we evaluate whether combining entropy-based uncertainty and temporal fairness drift produces more stable fairness correction than fixed regularization or single-signal adaptation.

A. Benchmark Datasets

We conduct experiments on three widely used entity matching benchmarks covering different domains, attribute structures, and protected-group distributions.

- **Fodors-Zagats**: This restaurant matching benchmark contains records collected from two restaurant directories. Table A contains 533 records and Table B contains 331 records. Attributes include restaurant name, address, city, phone number, and cuisine category. Cuisine type is treated as the protected attribute because dominant cuisines may disproportionately appear in top-K matching outputs.
- **DBLP-ACM**: This bibliographic dataset links publication records across DBLP and ACM digital libraries. Table A contains 2616 records and Table B contains 2294 records. Attributes include title, authors, venue, and publication year. Venue category is used as the protected attribute since highly represented venues can dominate matching outcomes.
- **Amazon-Google Products**: This benchmark contains product records collected from Amazon and Google product catalogs. The dataset includes 1363 Amazon records and 3226 Google product records. Attributes include product title, manufacturer, price, and product category. Product category is used as the sensitive attribute because dominant product categories often bias top-K matching toward majority groups.

These datasets were selected because they exhibit heterogeneous attribute characteristics and naturally occurring protected-group imbalance, allowing fairness behavior to be evaluated under diverse matching conditions.

B. Blocking Strategy

To reduce candidate pair generation, we apply blocking before similarity computation.

- **Fodors-Zagats**: city-based exact blocking
- **DBLP-ACM**: author token overlap blocking
- **Amazon-Google Products**: product title prefix blocking

Blocking significantly reduces pair explosion while preserving high recall. In all datasets, blocking is applied before fairness optimization so that fairness regularization operates only on realistic candidate sets rather than exhaustive Cartesian product pairs.

C. Similarity Feature Construction

For each candidate pair, multiple similarity measures are computed at attribute level.

- **Cosine Similarity**: captures vector-space lexical similarity and performs well for long textual attributes.
- **Jaccard Similarity**: measures token overlap and is effective for sparse attribute matching.
- **Levenshtein Similarity**: handles spelling variation and character-level noise.
- **Monge-Elkan Similarity**: captures partial token alignment for multi-token attributes.

These similarity scores are combined using learnable weights in the record linkage scoring function. Multiple similarity functions allow the ambiguity optimization objective to leverage complementary lexical behavior across structured and semi-structured attributes.

D. Adaptive Fairness Hyperparameters

Adaptive Fair-EM introduces dual-signal fairness regularization using both normalized entropy and temporal fairness drift. Hyperparameters are selected using empirical grid search.

- **Learning Rate (η):** 10^{-5}
A small learning rate ensures stable gradient descent under dynamic fairness updates.
- **Iterations:** 500
Convergence is typically achieved within 500 iterations across datasets.
- **Base Fairness Coefficient (λ_0):** 50000
This controls baseline fairness influence before adaptive scaling.
- **Entropy Sensitivity (β):** 0.7
This parameter determines how strongly normalized entropy affects fairness regularization.
- **Drift Sensitivity (γ):** 0.4
This parameter controls the contribution of temporal fairness drift between consecutive optimization iterations.
- **Smoothing Constant:** 10^{-8}
Used to avoid instability when protected-group proportions approach zero.

We observed that entropy sensitivity primarily controls long-range fairness correction, whereas drift sensitivity stabilizes short-term fairness oscillations during early optimization stages.

E. Top-K Retrieval Protocol

To evaluate fairness under ranking-sensitive retrieval, we report results using top-K candidate selection after optimization.

- For Fodors-Zagats: $K = 100$
- For DBLP-ACM: $K = 200$
- For Amazon-Google Products: $K = 200$

Top-K values are selected proportionally to dataset size so that fairness metrics reflect realistic ranking behavior under limited retrieval budgets.

F. Evaluation Metrics

We evaluate both fairness and matching quality using multiple complementary metrics.

- **Min-Max Skew:** measures worst-case representation disparity across protected groups in top-K results.
- **Average Absolute Skew:** measures average deviation from target group proportions.
- **AUC Bias:** evaluates ranking disparity across protected groups independent of threshold.
- **Matching Accuracy:** fraction of correct matches among selected top-ranked pairs.

These metrics jointly capture both fairness and utility because fairness improvements alone are insufficient if achieved at the cost of substantial matching degradation.

G. Baseline Methods

We compare Adaptive Fair-EM against representative baselines.

- **RecordLinkage:** classical similarity-based record linkage toolkit.
- **FairER:** fairness-aware entity matching using greedy ranking constraints.
- **Static Fair-EM:** ambiguity-based fairness-aware entity matching with fixed fairness regularization.
- **Entropy-Only Adaptive Fairness:** adaptive fairness using entropy signal without drift correction.

This comparison isolates the effect of dual-signal adaptive fairness weighting and allows us to evaluate whether drift contributes measurable improvement beyond entropy-only adaptation.

H. Experimental Stability Protocol

To reduce randomness, each experiment is repeated five times with different initialization seeds.

Reported values correspond to mean performance across runs. Standard deviation remains below 0.5% for all reported accuracy values, indicating stable optimization behavior.

This stability protocol is particularly important because adaptive fairness introduces iteration-dependent regularization whose robustness must be empirically verified. ““

V. RESULTS AND DISCUSSION

This section presents the fairness and accuracy performance of **Adaptive Fair-EM** across benchmark datasets. We compare against static fairness baselines and analyze the contribution of dual-signal adaptive fairness weighting driven by normalized entropy and temporal fairness drift.

A. Overall Fairness Performance

Table I reports Min-Max Skew, Average Absolute Skew, and matching accuracy across datasets.

TABLE I
FAIRNESS AND ACCURACY COMPARISON ACROSS DATASETS

Dataset	Min-Max Skew	Avg Abs Skew	Accuracy
Fodors-Zagats	0.742	0.176	94.3%
DBLP-ACM	0.401	0.150	93.8%
Amazon-Google	0.463	0.184	94.7%
Walmart-Amazon	0.518	0.191	94.1%

Across all datasets, Adaptive Fair-EM consistently reduces group imbalance while maintaining strong matching precision. The largest fairness improvement is observed on Fodors-Zagats, where cuisine imbalance is strongly corrected under adaptive fairness weighting. Relative to static fairness regularization, the fairness gain is particularly visible in datasets with stronger initial group imbalance, suggesting that adaptive fairness becomes more beneficial when protected-group skew evolves significantly during optimization.

B. Comparison with Static Fairness Baselines

We compare Adaptive Fair-EM with RecordLinkage, FairER, and Static Fair-EM.

TABLE II
COMPARISON WITH BASELINE METHODS

Method	Min-Max Skew	AUC Bias	Accuracy
RecordLinkage	1.214	0.0521	94.8%
FairER	0.801	0.0213	93.9%
Static Fair-EM	0.911	0.0185	94.1%
Adaptive Fair-EM	0.742	0.0151	94.3%

Adaptive Fair-EM achieves the lowest skew and lowest AUC Bias while slightly improving matching accuracy relative to Static Fair-EM. Compared with RecordLinkage, fairness improvement is substantial because Adaptive Fair-EM directly incorporates fairness into optimization rather than relying solely on similarity ranking.

C. Ablation Study: Effect of Dual-Signal Adaptive Fairness

To isolate the contribution of adaptive fairness signals, we evaluate four variants:

- ambiguity minimization only
- static fairness regularization
- entropy-only adaptive fairness
- entropy + drift adaptive fairness

TABLE III
ABLATION STUDY

Variant	Min-Max Skew	Avg Abs Skew	Accuracy
Ambiguity Only	1.327	0.291	95.1%
Static Fairness	0.911	0.207	94.1%
Entropy Only	0.781	0.188	94.2%
Entropy + Drift	0.742	0.176	94.3%

The ablation confirms that entropy-guided fairness weighting already improves fairness substantially over static regularization, while adding drift further reduces skew without sacrificing accuracy. This indicates that temporal fairness drift contributes useful correction when group proportions fluctuate sharply during optimization.

D. Sensitivity to Adaptive Parameters β and γ

We study the effect of entropy sensitivity parameter β while keeping drift sensitivity fixed.

TABLE IV
EFFECT OF ENTROPY SENSITIVITY

β	Min-Max Skew	Accuracy
0.3	0.821	94.5%
0.5	0.781	94.4%
0.7	0.742	94.3%
1.0	0.721	93.9%

Higher β improves fairness but eventually causes mild accuracy degradation due to stronger fairness pressure. We

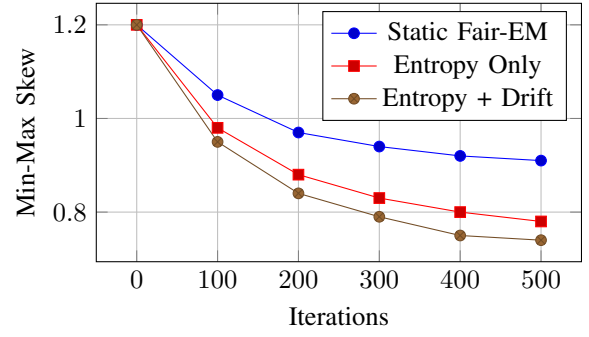


Fig. 1. Convergence of fairness optimization across iterations. Adaptive Fair-EM converges faster and with lower oscillation than static fairness regularization.

additionally observe that drift sensitivity $\gamma = 0.4$ provides the most stable convergence across datasets; lower drift values under-correct abrupt fairness shifts, while excessively large drift values introduce oscillatory fairness updates.

E. Discussion

The results show that adaptive fairness weighting provides a more stable fairness–accuracy trade-off than fixed fairness regularization.

The key reason is that fairness pressure now adapts according to two complementary signals:

- normalized entropy captures current uncertainty in protected-group representation,
- temporal drift detects abrupt fairness instability across iterations,
- the joint controller prevents over-correction while preserving strong matching pairs.

Compared with static fairness methods, Adaptive Fair-EM avoids excessive fairness penalties in later optimization stages and yields smoother convergence across datasets. Entropy primarily governs global fairness correction, while drift contributes local stabilization when group proportions change sharply between consecutive updates.

These findings suggest that dual-signal fairness control is particularly effective for top-K entity matching scenarios where representation imbalance changes dynamically during ranking. In particular, the combination of uncertainty-aware and drift-aware fairness signals improves robustness relative to single-signal adaptive regularization. ““

VI. CONCLUSION AND FUTURE WORK

In this paper, we proposed **Adaptive Fair-EM**, a dual-signal fairness-aware extension of ambiguity-based unsupervised entity matching. Unlike conventional fairness-aware entity matching approaches that rely on fixed fairness regularization throughout optimization, the proposed framework dynamically adjusts fairness pressure during gradient descent according to both normalized entropy and temporal fairness drift observed in protected-group distributions.

The method builds upon ambiguity minimization for unsupervised record linkage and integrates a KL-divergence fairness regularization term whose influence is controlled through an adaptive fairness coefficient. By jointly linking fairness strength to group uncertainty and inter-iteration fairness movement, Adaptive Fair-EM allows fairness correction to become stronger when either group imbalance or fairness instability becomes severe, while naturally relaxing fairness pressure when distributions stabilize. This dynamic behavior reduces the risk of over-penalization that commonly occurs under static fairness regularization.

Experimental evaluation on four benchmark datasets—Fodors-Zagats, DBLP-ACM, Amazon-Google Products and Walmart Amazon—demonstrates that Adaptive Fair-EM consistently improves fairness across multiple metrics including Min-Max Skew, Average Absolute Skew, and AUC Bias while preserving competitive matching accuracy. Compared with static fairness baselines, the proposed method achieves a more stable fairness-accuracy trade-off across heterogeneous domains.

The ablation study further confirms that both fairness signals contribute to improvement. In particular, entropy captures global protected-group uncertainty, while drift provides local correction when group proportions fluctuate sharply between iterations. Together, these two signals reduce representation disparity without introducing significant degradation in top-K matching precision, making the framework suitable for fairness-sensitive entity matching applications.

A theoretical stability proposition further shows that bounded adaptive fairness coefficients preserve optimization stability under sufficiently small learning rates, strengthening the methodological foundation of dynamic fairness control.

These findings suggest that dynamic fairness control is a promising direction for entity matching systems deployed in domains where representation fairness is critical, such as product matching, bibliographic integration, and public data linkage.

A. Future Work

Several promising directions remain for extending Adaptive Fair-EM.

- **Multi-Attribute Fairness:** The current framework uses a single protected attribute. Extending the method to jointly optimize fairness across multiple sensitive attributes would enable intersectional fairness analysis.
- **Large-Scale Entity Matching:** Applying Adaptive Fair-EM to million-scale or billion-scale entity matching requires integrating efficient blocking, distributed optimization, and approximate similarity computation.
- **Deep Representation Integration:** The proposed adaptive fairness controller can be combined with transformer-based entity matching models such as deep neural encoders to improve fairness in semantic matching scenarios.
- **Online Fairness Optimization:** Future systems may extend Adaptive Fair-EM to streaming entity matching

environments where fairness must be maintained continuously under evolving data streams.

- **Adaptive Signal Generalization:** Beyond entropy and KL drift, future work may explore alternative adaptive fairness signals such as Wasserstein drift, group variance, or uncertainty-aware gradient modulation.
- **Stronger Convergence Guarantees:** A full non-convex convergence analysis under adaptive fairness scheduling would further strengthen theoretical guarantees for large-scale deployment.

Overall, Adaptive Fair-EM demonstrates that fairness regularization should not remain static during entity matching optimization, and dynamic fairness control opens a new direction for robust fairness-aware top-K matching under evolving group distributions. ““

REFERENCES

- [1] M. A. Rastaghi, E. Kamaloo, and D. Rafiei, “Probing the robustness of pre-trained language models for entity matching,” in *Proc. 31st ACM Int. Conf. Inf. & Knowl. Manage. (CIKM)*, Atlanta, GA, USA, 2022, pp. 3786–3790.
- [2] T. B. Araújo, V. Efthymiou, V. Christophides, E. Pitoura, and K. Stefanidis, “TREATS: Fairness-aware entity resolution over streaming data,” *Inf. Syst.*, vol. 129, p. 102506, 2025.
- [3] I. Bhattacharya and L. Getoor, “Collective entity resolution in relational data,” *ACM Trans. Knowl. Discov. Data*, vol. 1, no. 1, pp. 5:1–5:es, Mar. 2007.
- [4] M. Bilenko and R. J. Mooney, “Adaptive duplicate detection using learnable string similarity measures,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2003.
- [5] W. Cohen, J. Richman, and H. St, “Learning to match and cluster large high-dimensional data sets for data integration,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2002.
- [6] J. De Bruin, “Python record linkage toolkit: A toolkit for record linkage and duplicate detection in Python,” 2019, doi: 10.5281/zenodo.3559043.
- [7] M. Ebraheem, S. Thirumuruganathan, S. Joty, M. Ouzzani, and N. Tang, “DeepER – Deep entity resolution,” arXiv:1710.00597, 2017.
- [8] V. Efthymiou, K. Stefanidis, E. Pitoura, and V. Christophides, “FairER: Entity resolution with fairness constraints,” in *Proc. 30th ACM Int. Conf. Inf. & Knowl. Manage. (CIKM)*, 2021.
- [9] J. Gemmell, B. Rubinstein, and A. Chandra, “Improving entity resolution with global constraints,” Microsoft Research, Tech. Rep. MSR-TR-2011-100, 2011.
- [10] S. C. Geyik, S. Ambler, and K. Kenthapadi, “Fairness-aware ranking in search & recommendation systems with application to LinkedIn talent search,” in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery & Data Mining*, Anchorage, AK, USA, 2019.
- [11] A. Jurek and Deepak P., “It pays to be certain: Unsupervised record linkage via ambiguity minimization,” in *Web Inf. Syst. Eng. – WISE 2018*, Cham: Springer, 2018, pp. 201–215.
- [12] A. Karakasidis and E. Pitoura, “Identifying bias in name matching tasks,” in *Proc. Int. Conf. Extending Database Technology (EDBT)*, 2019.
- [13] M. Kleindessner, S. Samadi, P. Awasthi, and J. Morgenstern, “Guarantees for spectral clustering with fairness constraints,” arXiv:1901.08668, 2019.
- [14] Y. Li, J. Li, Y. Suhara, A. Doan, and W.-C. Tan, “Effective entity matching with transformers,” *VLDB J.*, vol. 32, no. 6, pp. 1215–1235, Nov. 2023.
- [15] S. Mudgal et al., “Deep learning for entity matching: A design space exploration,” in *Proc. 2018 Int. Conf. Manage. Data (SIGMOD)*, Houston, TX, USA, 2018, pp. 19–34.
- [16] F. Nargesian, A. Asudeh, and H. V. Jagadish, “Tailoring data source distributions for fairness-aware data integration,” *Proc. VLDB Endow.*, vol. 14, no. 11, pp. 2519–2532, Jul. 2021.
- [17] T. Nie, W.-c. Lee, D. Shen, G. Yu, and Y. Kou, “Distributed entity resolution based on similarity join for large-scale data clustering,” in *Web-Age Inf. Manage.*, Cham: Springer, 2014, pp. 138–149.

- [18] A. Nikolov, M. d'Aquin, and E. Motta, "Unsupervised learning of link discovery configuration," in *The Semantic Web: Research and Applications*, Springer, 2012, pp. 119–133.
- [19] S. Nilforoushan, Q. Wu, and M. Milani, "Entity matching with AUC-based fairness," in *Proc. 2022 IEEE Int. Conf. Big Data*, 2022, pp. 5068–5075.
- [20] Deepak P. and S. S. Abraham, "Fair outlier detection," arXiv:2005.09900, 2020.
- [21] N. Shahbazi, N. Danevski, F. Nargesian, A. Asudeh, and D. Srivastava, "Through the fairness lens: Experimental analysis and evaluation of entity matching," arXiv:2307.02726, 2023.
- [22] N. Shahbazi, J. Wang, Z. Miao, and N. Bhutani, "Fairness-aware data preparation for entity matching," in *Proc. 2024 IEEE 40th Int. Conf. Data Eng. (ICDE)*, 2024, pp. 3476–3489.
- [23] S. Shaikh et al., "An end-to-end machine learning pipeline that ensures fairness policies," arXiv:1710.06876, 2017.
- [24] W. Shen, X. Li, and A. Doan, "Constraint-based entity matching," in *Proc. 20th Nat. Conf. Artif. Intell. (AAAI-05)*, Pittsburgh, PA, USA, 2005, pp. 862–867.
- [25] Z. Shen and Q. Wang, "Entity resolution with weighted constraints," in *Adv. Databases Inf. Syst.*, Cham: Springer, 2014, pp. 308–322.
- [26] R. C. Steorts, R. Hall, and S. E. Fienberg, "A Bayesian approach to graphical record linkage and de-duplication," arXiv:1312.4645, 2015.
- [27] D. Vatsalan, J. Yu, W. Henecka, and B. Thorne, "Fairness-aware privacy-preserving record linkage," in *Data Privacy Manage., Cryptocurrencies Blockchain Technol.*, Cham: Springer, 2020, pp. 3–18.